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TrustRAG: A Trustworthy Offline Retrieval-Augmented Generation Framework with Explainable Multi-Signal Hallucination Detection

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ABSTRACT Large Language Models (LLMs) have demonstrated remarkable capabilities across knowledge-intensive applications; however, they remain susceptible to hallucinations, producing fluent yet factually incorrect responses that undermine reliability in real-world deployments. Although Retrieval-Augmented Generation (RAG) improves factual grounding through external knowledge retrieval, existing approaches often provide limited explainability, weak evidence attribution, and insufficient verification mechanisms, particularly in offline and privacy-sensitive environments. This paper proposes TrustRAG, a trustworthy offline Retrieval-Augmented Generation framework that integrates dense semantic retrieval, explainable evidence attribution, and multi-signal hallucination detection within a unified response validation pipeline. The framework employs Sentence Transformer embeddings and a local FAISS vector database for semantic retrieval, followed by an intrinsic explainability module that performs provenance tracking and cosine-based evidence attribution. Generated responses are subsequently verified using three complementary signals: Natural Language Inference (NLI), BERTScore semantic similarity, and embedding space cosine similarity. These verification signals are combined into a unified Trust Score that drives a tri-state decision controller for classifying generated sentences as Supported, Weakly Supported, or Hallucinated, thereby enabling automatic response validation and evidence-aware decision making. Experimental evaluation on an offline benchmark demonstrates that TrustRAG achieves an F1-score of 0.898 and an AUC-ROC of 0.951, substantially reducing hallucinated responses while improving provenance tracking, semantic consistency, and overall trustworthiness compared with a conventional RAG pipeline. The framework provides a privacy-preserving solution for deploying explainable and trustworthy LLM-based systems in enterprise, healthcare, legal, and other security-sensitive applications.

INDEX TERMS Explainable Artificial Intelligence (XAI), Hallucination Detection, Large Language Models (LLMs), Natural Language Inference, Offline Retrieval-Augmented Generation (RAG), Trustworthy Retrieval-Augmented Generation, Trust Score.

I. INTRODUCTION

Large Language Models (LLMs) have revolutionized natural language processing (NLP) by demonstrating remarkable capabilities in language understanding, reasoning, and text generation across a wide range of knowledge-intensive tasks. The advent of Transformer based architectures, followed by advances in large-scale pre-training and instruction tuning, has significantly improved the performance of LLMs in applications such as question answering, conversational assistants, document summarization, code generation, and enterprise knowledge management [1], [19], [20], [23]. Owing to their ability to generate coherent and contextually relevant responses, LLMs are increasingly being integrated into real-world decision support systems spanning healthcare, finance, education, legal services, and scientific research.

However, even with all these improvements, there are still

difficulties concerning the factual accuracy of the outputs provided by LLMs. As LLMs predict the probability of the next token and not the factual veracity of the output, their responses may contain misleading, contradictory, or false information even though they sound coherent and well-informed. This problem is known as *hallucination* and greatly constrains the applicability of LLMs in safety-sensitive tasks that require factual accuracy and user trust [2], [3]. Several researchers found that hallucinations can emerge due to the lack of parametric knowledge, contextual information, or reasoning ability of an LLM.

Retrieval Augmented Generation (RAG) is one of the most efficient ways to address the issue of hallucination in the generation task, which involves augmenting language models with external knowledge sources when performing inference tasks. While conventional methods rely on the knowledge stored inside the parameters of the model, RAG retrieves semantically relevant documents from an external dataset and uses this information in the generation step. Fundamental models that use the idea of retrieval augmentation, like RAG, REALM, and Dense Passage Retrieval (DPR), have shown the efficiency of language generation with the help of retrieval, and dense embeddings like Sentence-BERT and efficient vector indexing systems like FAISS have provided scalability for this purpose [4]–[6], [21], [22].

With the rising usage of RAG, the implementation of RAG in specialized environments, especially those that demand trustworthy decision-making processes, has been boosted. Systematic reviews carried out recently have revealed that the usage of RAG in knowledge-intensive applications, such as healthcare, is increasingly becoming common; however, various issues have been noted with regard to the quality of the retrieval process, attribution of evidence, validation of facts, and reliability of responses [30].

Although RAG substantially improves factual consistency compared with standalone LLMs, retrieval alone cannot completely eliminate hallucinations. In many cases, language models continue to generate unsupported statements when the retrieved evidence is incomplete, ambiguous, only partially relevant, or ignored during response generation. Moreover, most existing RAG systems primarily emphasize retrieval effectiveness while providing limited mechanisms for explaining why a response should be trusted or for verifying whether the generated content is fully supported by the retrieved evidence. As a result, users often receive fluent responses without transparent evidence attribution or an explicit measure of response reliability, limiting the practical deployment of RAG systems in high-assurance applications [7]–[10].

Recent research has therefore shifted from improving retrieval quality alone toward developing reliable hallucination detection and response verification techniques for Retrieval-Augmented Generation systems. Existing approaches investigate complementary strategies including self-consistency verification, factual consistency evaluation, natural language inference (NLI), semantic similarity analysis, uncertainty estimation, and evidence-grounded reasoning to identify unsupported or fabricated responses generated by LLMs [14]–[18]. More recent studies have further explored reinforcement learning, logical reasoning, entity verification, uncertainty quantification, and benchmark-driven evaluation for improving hallucination detection in RAG pipelines [24]–[29]. These works have greatly contributed to the state-of-the-art and highlight the increasing role of response validation in contemporary LLM-based systems.

At the same time, Explainable Artificial Intelligence (XAI) has become a crucial factor that helps to increase the transparency and accountability of intelligent systems. Popular model agnostic methods of interpretation such as Local Interpretable Model Agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP) offer a range of information about features' importance and predictions made by the model [11], [12]. Nevertheless, these methods are mainly focused on traditional prediction models and do not provide any explanation of the relationship between retrieved information and a generated answer in Retrieval Augmented Generation systems. As a result, users obtain a generated answer without any knowledge about which information sources served as evidence for generating the answer, the strength of correspondence between evidence and generated answer, or whether the answer can be considered credible.

While prior work has studied each of retrieval enhancement, hallucination detection, factual consistency checking, and explainability independently, only a handful of approaches have addressed all four of these complementary factors in an integrated framework that can provide semantic search, evidence attribution, multisignal verification, explainable assistance, and trust quantification at the same time. Additionally, most of these existing approaches use cloud architectures, making them unsuitable for deployment in scenarios that require local processing. This leads to the motivation of developing a framework that integrates knowledge retrieval, evidence attribution, and hallucination detection, among others, in a secure offline setting.

To overcome these difficulties, TrustRAG has been developed as a trustworthy offline Retrieval-Augmented Generation framework with integration of dense semantic retrieval, explainable evidence attribution, and multi-signal hallucination detection into a single response validation pipeline. The TrustRAG framework leverages semantic retrieval based on Sentence-BERT and local vector database using FAISS. To improve response reliability, TrustRAG jointly exploits Natural Language Inference (NLI), BERTScore semantic similarity, and embedding-space cosine similarity to compute a unified Trust Score that supports evidence aware response validation through a tri-state decision mechanism. This is made possible by incorporating all three elements in one unified architecture. This would help improve the factual accuracy of the system while also giving transparent and interpretable decision support systems in offline settings.

The principal contributions of this paper are summarized as follows:

1) The trustworthy offline Retrieval-Augmented Genera-

tion system, known as TrustRAG, combines semantic retrieval, explainable evidence attribution, and multi-signal hallucination detection in a response validation pipeline.
2) The Multi-Signal Trust Score is a combination of NLI score, BERTScore, and cosine similarity scores that provides reliable hallucination detection and response validation.
3) The intrinsic explainability module supports provenance tracking and evidence attribution and, therefore, increases the explainability of generated responses.
4) The trust-aware tri-state decision system classifies generated sentences into categories of Supported, Weakly Supported, or Hallucinated sentences, providing the sentence-level assessment and the response-level control of reliability.
5) The experiments have shown that the system is superior in hallucination detection, provenance tracking, semantic consistency, and response reliability to the baseline Retrieval-Augmented Generation system.

The rest of this paper is organized in the following manner. Section II deals with the literature review of Retrieval-Augmented Generation, hallucination detection, Explainable Artificial Intelligence, and trust-worthy Large Language Models. Section III introduces the proposed framework of TrustRAG along with its architecture. Section IV focuses on the methodology of experimentation along with its implementation and performance evaluation metrics. Section V presents the experimental results and comparative analysis while Section VI concludes the paper.

II. RELATED WORK

A. LARGE LANGUAGE MODELS AND RETRIEVAL-AUGMENTED GENERATION

Rapid progress of Large Language Models (LLMs) has contributed greatly to Natural Language Processing (NLP) through the realization of significant advances in language comprehension, reasoning, and text generation. The transformer architecture marked the start of the current development trend in language modeling with the introduction of self-attention that can be used to model long-distance contextual relationships [20]. Pre-trained language models like BERT proved the value of bidirectional context embeddings in downstream NLP tasks, and large-scale autoregressive language models extended the scope of text generation tasks with pre-training on vast amounts of text data [1], [19]. More recently, instruction-tuned LLMs, in particular the Llama series, achieved considerable success in reasoning, dialogue generation, and knowledge-based question answering tasks [23].

Although there have been incredible breakthroughs in the performance of these language models, the basic limitation for the LLMs still lies in the information embedded in the model parameters. Because the information embedded in these models is not self-updating after training, these models will not be able to use new information without being trained further. Also, they often have trouble dealing with domain information, dynamic information, and fact-checking because of which they produce old or even wrong information in some cases [2], [3].

Retrieval-Augmented Generation (RAG) has been established as a successful methodology in enhancing the factual grounding of large language models (LLMs) using document retrieval before responding. As opposed to being entirely dependent on parametric knowledge, the methodology of RAG uses neural search along with generative language models where the generated response can be conditioned upon retrieved evidences [4]. The methodologies such as REALM and Dense Passage Retrieval (DPR) were developed later that enhanced the accuracy of the retrieval process by utilizing dense vector representations and end-to-end training [5], [6].

New trends in the field have helped enhance retrieval efficiency using semantic embeddings and vector indexing techniques. Sentence-BERT can be used to create semantic embeddings of sentences, which in turn help to greatly boost the performance of dense retrieval, whereas FAISS helps to conduct scalable approximate nearest neighbor searches from large collections of documents [21], [22]. This has enabled the rapid deployment of RAG in several applications ranging from enterprise knowledge management to customer service and even healthcare [30].

Despite the fact that retrieval significantly increases consistency compared to standalone language models, retrieval cannot completely prevent hallucinations. Quality of the responses generated depends on the efficiency of the retrieval process, relevance of the documents retrieved, and completeness of the evidence contained in the documents. The documents retrieved might include incomplete or inconsistent data, thus leading to unsupported statements made by language models, even though they successfully retrieve relevant documents. In addition to that, currently used RAG systems focus on the accuracy of the retrieval process and the quality of the response generated, but provide minimal opportunities for evidence attribution, explanation of reasoning, and quantifying the level of trust [7], [8], [10].

It can be concluded that although RAG has greatly increased the quality of the knowledge-grounded text generation, verification of the response generated and explanation of evidence attribution remain unresolved research problems. These limitations need to be solved by combining the retrieval process with trusted verification mechanisms that will be able to evaluate both consistency and reliability of the evidence.

B. HALLUCINATION DETECTION IN LARGE LANGUAGE MODELS

Growing use of Large Language Models (LLMs) has spurred research on detection of hallucinations and factual consistency verification, especially in RAG settings. Although retrieval can improve factual grounding, language models could produce responses containing unfounded, made-up or illogical information even if they have access to the supporting facts. Thus, accurate detection of hallucinations has emerged

as a crucial step towards reliable deployment of LLMs for knowledge-heavy applications.

Early works mostly concentrated on measuring factual consistency via Natural Language Inference (NLI) and semantic similarity approaches. FactCC pioneered in developing a weakly supervised approach for detecting factual inconsistencies in abstractive summarization by discovering contradictions between summaries and their sources [14]. Afterward, SummaC showed that reasoning through NLI helps to achieve much higher inconsistency detection performance through semantic entailment modeling between source documents and summaries [15]. TRUE benchmark, in turn, has presented a comprehensive evaluation framework for measuring factual consistency in various text generation tasks, revealing the shortcomings of the existing metrics in different generation scenarios [16].

Several state-of-the-art advances have emerged that propose new techniques tailored to LLMs' hallucination detection. SelfCheckGPT is a technique for zero-resource hallucination detection that leverages consistency of generated outputs through analysis of multiple independently generated outputs without any use of external knowledge sources [8]. FactScore is a metric that measures factual accuracy at the level of atomic facts and thus allows for detailed assessment of long form-generated texts [9]. In addition, RAGAS is an automated evaluation method for Retrieval-Augmented Generation systems that simultaneously evaluates the quality of retrieval, answer correctness, faithfulness, and context-awareness without manual annotation of reference answers [10]. Despite all that, the recent developments have mostly emphasized evaluation and post-generation verification aspects rather than integration of multiple verification signals into the end-to-end response generation pipeline.

With the rapid development of Retrieval-Augmented Generation methods, researchers started paying more attention to evidence-aware hallucination detection. In particular, several works recently explored various techniques such as reasoning-based verification, entity-level evidence alignment, logical consistency analysis, uncertainty estimation, and benchmark studies. RLSeek proposes reinforcement learning for evidence-grounded reasoning for hallucination detection which enables fact verification based on iterative reasoning with respect to retrieved documents [24]. Evidence-Aligned Entity Verification improves factuality verification by performing explicit validation of generated entities against the retrieved evidence [25]. Logic Matters shows how logical reasoning plays a crucial role in improving lightweight hallucination classification in Retrieval-Augmented Generation pipelines [27] while MARCH utilizes multi-agent collaborative verification to improve response faithfulness based on cooperative evidence verification [28]. In addition, Faithfulness-Aware Uncertainty Quantification provides estimation of confidence for generated responses taking into account both factual faithfulness and predictive uncertainty [29]. Lastly, comprehensive benchmarking studies have shown that existing methods of hallucination evaluation are not enough for reliable factual verification in Retrieval-Augmented Generation settings [26].

However, despite all these advances, there are still some issues that have not yet been addressed. First of all, most existing solutions focus on analyzing hallucinations, verifying evidence, evaluating retrieval quality, or estimating uncertainty as separate processes without including them in a comprehensive framework of TRAG. In addition to this, many approaches are primarily focused on detecting hallucinations but do not provide any evidence attribution, explainability of decision making, and quantification of the generated response trustworthiness. All this leads to a lack of trust of users, especially in sensitive contexts where reliability of response is equally important as its trustworthiness.

C. EXPLAINABLE ARTIFICIAL INTELLIGENCE FOR TRUSTWORTHY LARGE LANGUAGE MODELS

Explainable Artificial Intelligence (XAI) is currently a primary research direction in creating transparent, interpretable, and accountable machine learning systems. Given the growing use of artificial intelligence in crucial applications, users demand explanations to understand the rationale behind the model's predictions and increase the reliability of automation. Local Interpretable Model-Agnostic Explanations (LIME) is one of the most popular model-agnostic explanation methods that locally approximate the decision boundaries of the model and reveal the most important input features, while SHapley Additive exPlanations (SHAP) leverages game theory concepts to quantify the importance of each input feature [11], [12]. The methods have already found wide use in numerous classification and regression tasks to increase model interpretability and user trust.

Existing XAI approaches offer great explanatory power when it comes to predicting using ML models but face difficulties when it comes to explaining retrieval-augmented generation (RAG). In contrast to traditional supervised learning algorithms where the user needs explanations about the prediction, the RAG system has many interlinked components, such as document retrieval, semantic matching, response generation, and evidence incorporation. Therefore, providing an explanation for the prediction is insufficient because the relationship between the retrieved document and the generated response should be justified.

These findings suggest that explainability needs to be viewed as an integral part of trustworthiness of RAG, rather than just another post-hoc interpretation tool. Specifically, good explainability for RAG has to define clear connections between the retrieved evidence and the generated output, while enabling facts verification and response-level reliability estimation. At the same time, a number of currently available solutions keep viewing explainability as a post-hoc procedure and not as an integral part of the process of response generation and verification. Moreover, explainability is often evaluated in isolation from hallucination detection, which complicates the concurrent verification of the output facts, its evidential support, and reliability. All these observations

TABLE 1. Comparison of Representative Approaches for Trustworthy Retrieval-Augmented Generation

Representative Approach	Dense Semantic Retrieval	Explainable Evidence Attribution	Multi-Signal Hallucination Detection	Offline/ Local Execution	Trust Score
Lewis *et al.* [4]	✓	×	×	×	×
REALM [5]	✓	×	×	×	×
Dense Passage Retrieval (DPR) [6]	✓	×	×	×	×
LIME / SHAP [11], [12]	×	△	×	×	×
SelfCheckGPT [8]	×	×	✓	✓	×
FactScore [9]	×	×	✓	✓	×
RAGAS [10]	✓	△	✓	✓	×
RLSeek [24]	✓	△	✓	×	×
Evidence-Aligned Entity Verification [25]	✓	△	✓	×	×
MARCH [28]	✓	×	✓	×	×
TrustRAG (Proposed)	✓	✓	✓	✓	✓

Note: ✓ = Fully supported; △ = Partially supported; × = Not supported.

make it necessary to develop explainability mechanisms that would work in combination with retrieval and verification rather than be separate interpretability tools.

These observations indicate that for trustworthy RAG it is not sufficient to have reliable retrieval and effective hallucination detection; it is also necessary to have explainability that allows understanding how exactly the generated response can be trusted.

D. RESEARCH GAP AND MOTIVATION

In light of such findings, the proposed framework TrustRAG employs semantic retrieval, explainable evidence attribution, multi-modal hallucination detection, and trust-aware response validation. Unlike approaches that independently address retrieval, explainability, or hallucination detection, the framework combines these complementary capabilities to improve factual reliability while providing transparent evidence support and quantitative trust assessment. Existing studies have substantially improved semantic retrieval, factual consistency evaluation, evidence verification, and response assessment through complementary techniques including dense retrieval, Natural Language Inference (NLI), semantic similarity, uncertainty estimation, and reinforcement learning. However, these approaches primarily address individual aspects of trustworthy response generation rather than providing a unified framework that simultaneously integrates semantic retrieval, explainable evidence attribution, multi-signal hallucination detection, quantitative trust assessment, and offline deployment. As a result, ensuring transparent, reliable, and trustworthy response generation remains an open research challenge, particularly in privacy-sensitive environments where external cloud-based verification cannot be employed.

The salient features of popular Retrieval-Augmented Generation models are summarized in Table 1. Despite the proven success of existing techniques in improving retrieval capabilities, fact checking, and hallucination detection, current efforts are predominantly aimed at addressing individual aspects of the response generation process. Specifically, explainability is often considered as an after-the-fact study, not as an inherent element of the response validation, while the quantitative estimation of trustworthiness is hardly ever taken into account. Furthermore, many recent approaches rely on cloud-based infrastructures, limiting their applicability in security-sensitive environments requiring local inference and strict data confidentiality.

Motivated by these observations, this work introduces **TrustRAG**, a trustworthy offline Retrieval-Augmented Generation framework that integrates semantic retrieval, explainable evidence attribution, multi-signal hallucination detection, and trust-aware response validation within a unified architecture. Unlike existing approaches that independently address retrieval, explainability, or hallucination detection, the proposed framework combines these complementary capabilities to improve factual reliability while providing transparent evidence support and quantitative trust assessment. Consequently, TrustRAG offers a practical and interpretable solution for deploying trustworthy LLM-based systems in enterprise, healthcare, legal, and other privacy-sensitive applications.

III. PROPOSED TRUSTRAG FRAMEWORK

A. SYSTEM OVERVIEW

The proposed TrustRAG framework is designed to improve the factual reliability, transparency, and trustworthiness of Retrieval-Augmented Generation (RAG) systems operating in offline and privacy-sensitive environments. Unlike conventional RAG pipelines that primarily focus on document retrieval and response generation, TrustRAG integrates semantic retrieval, explainable evidence attribution, multi-signal hallucination detection, and trust-aware response validation within a unified architecture. The framework enables generated responses to be systematically verified against retrieved evidence before being presented to the user, thereby reducing hallucinations while improving interpretability and decision confidence.

Figure 1 illustrates the Dense Semantic Retrieval Module of the TrustRAG framework. Given a user query, the

framework first performs semantic document retrieval using Sentence-BERT embeddings and a local FAISS vector database to identify the most relevant supporting documents. The retrieved evidence is subsequently incorporated into the language model prompt to generate an evidence-grounded response while maintaining complete offline execution without reliance on external cloud services.

To improve transparency, an explainable evidence attribution module identifies the retrieved documents that contribute to each generated sentence through provenance tracking and cosine-based evidence alignment. This enables users to understand the relationship between retrieved evidence and generated content, thereby improving interpretability and facilitating manual verification when necessary.

Following response generation, TrustRAG performs hallucination detection using three complementary verification signals. Natural Language Inference (NLI) evaluates logical consistency between the retrieved evidence and each generated sentence, BERTScore measures contextual semantic similarity, and embedding-space cosine similarity quantifies semantic alignment between the generated sentence and the retrieved evidence. These complementary verification signals provide a multi-dimensional assessment of sentence-level response reliability.

The outputs of the verification modules are integrated into a sentence-level Trust Score. Based on predefined confidence thresholds, the Tri-State Decision Controller classifies each generated sentence as *Supported*, *Weakly Supported*, or *Hallucinated*. The sentence-level decisions are subsequently aggregated at the response level to determine whether the complete response should be retained or suppressed.

At the response level, the hallucination ratio is computed from the number of generated sentences classified as Hallucinated. If the hallucination ratio exceeds the predefined suppression threshold, the generated response is suppressed and the verified supporting evidence is returned instead. Otherwise, the response is retained together with its provenance information and supporting evidence. This hierarchical verification mechanism enables TrustRAG to provide both sentence-level reliability assessment and response-level safety control.

The proposed framework is intended for knowledge-intensive applications such as enterprise information systems, healthcare, legal services, and other privacy-sensitive environments where factual reliability, evidence traceability, and transparent decision support are important requirements.

B. DENSE SEMANTIC RETRIEVAL MODULE

The first stage of the TrustRAG framework retrieves semantically relevant documents that provide factual evidence for answering the user query. Unlike conventional keyword-based retrieval methods, semantic retrieval captures contextual meaning by representing both queries and documents in a shared dense embedding space. This enables the retrieval module to identify conceptually related documents even when lexical overlap between the query and document is limited, thereby improving retrieval robustness and reducing the likelihood of knowledge omission.

Let the user query be represented as

$$q = \{w_1, w_2, \ldots, w_n\}, \tag{1}$$

where w_i denotes the ith token in the input query.

Let the knowledge repository be represented as

$$D = \{d_1, d_2, \ldots, d_N\}, \tag{2}$$

where d_i denotes the ith indexed document chunk and N denotes the total number of document chunks stored in the local knowledge repository.

Both the query and documents are transformed into dense semantic embeddings using the Sentence-BERT encoder. The corresponding embedding representations are obtained as

$$e_q = f_{\mathrm{SBERT}}(q), \tag{3}$$

and

$$e_{d_i} = f_{\mathrm{SBERT}}(d_i), \tag{4}$$

where $f_{\mathrm{SBERT}}(\cdot)$ denotes the Sentence-BERT embedding function, $e_q \in \mathbb{R}^m$ represents the query embedding, and $e_{d_i} \in \mathbb{R}^m$ denotes the embedding of the ith document chunk.

Semantic similarity between the query and each document is computed using cosine similarity as

$$S(q, d_i) = \frac{e_q \cdot e_{d_i}}{\|e_q\| \|e_{d_i}\|}. \tag{5}$$

The cosine similarity score ranges from -1 to 1, where larger values indicate stronger semantic relevance between the query and the corresponding document.

To efficiently retrieve the most relevant evidence from large document repositories, all document embeddings are indexed using Facebook AI Similarity Search (FAISS), which performs approximate-nearest-neighbor search in the embedding space. Given the similarity scores, the retrieval module returns the top-k document chunks according to

$$\mathcal{R} = \mathrm{TopK}\left(S(q, d_i)\right), \tag{6}$$

where $\mathcal{R}$ denotes the retrieved evidence set containing the k most semantically relevant document chunks.

The retrieved evidence is subsequently incorporated into the prompt construction stage, where the language model generates an evidence-grounded response conditioned on both the user query and the retrieved contextual information. By employing dense semantic retrieval instead of conventional lexical matching, the framework is designed to improve contextual relevance while reducing retrieval failures caused by vocabulary mismatch. Furthermore, the use of a locally deployed FAISS vector database enables efficient offline retrieval, thereby supporting privacy-preserving deployment in enterprise and security-sensitive environments.

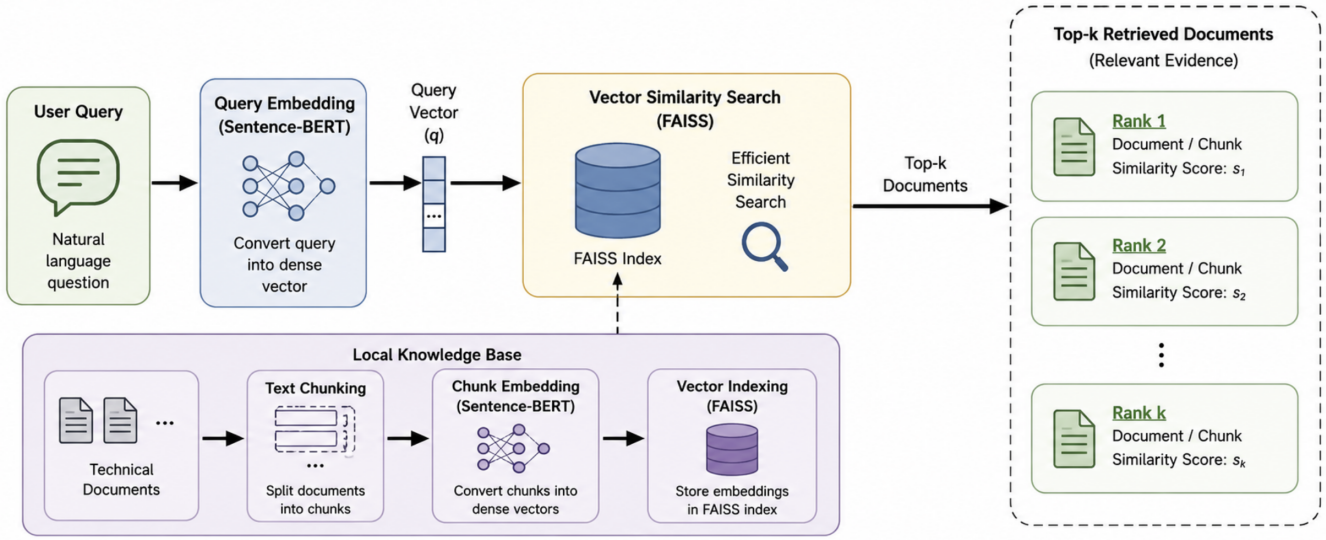


FIGURE 1. **Workflow of the proposed Dense Semantic Retrieval Module in the TrustRAG framework.**

As shown in Fig. 1, the Dense Semantic Retrieval Module transforms the user query into a dense semantic embedding using Sentence-BERT. The generated embedding is matched against the document embeddings stored in the local FAISS vector database through cosine similarity, and the top-k most relevant documents are retrieved for subsequent evidence attribution and response verification.

C. EXPLAINABLE EVIDENCE ATTRIBUTION MODULE

Following semantic retrieval, the proposed TrustRAG framework performs explainable evidence attribution to establish a transparent relationship between the retrieved documents and the generated content. Unlike conventional Retrieval-Augmented Generation (RAG) systems that present only the final response, the proposed module enables users to identify the supporting evidence associated with each generated sentence. This improves transparency and facilitates manual verification, particularly in applications requiring reliable and interpretable decision support.

Let the retrieved evidence set obtained from the Dense Semantic Retrieval Module be represented as

$$\mathcal{R} = \{r_1, r_2, \ldots, r_k\}, \tag{7}$$

where r_i denotes the ith retrieved document and k represents the number of retrieved documents.

The language model generates a response conditioned on the user query and the retrieved evidence as

$$G = f_{\mathrm{LLM}}(q, \mathcal{R}), \tag{8}$$

where q denotes the user query, $\mathcal{R}$ represents the retrieved evidence set, and $f_{\mathrm{LLM}}(\cdot)$ denotes the language model responsible for response generation.

Since the generated response may contain multiple factual statements, the response is decomposed into m individual sentences as

$$G = \{g_1, g_2, \ldots, g_m\}, \tag{9}$$

where g_j denotes the jth generated sentence.

To quantify the semantic contribution of each retrieved document toward each generated sentence, Sentence-BERT is used to obtain dense semantic representations. For the jth generated sentence, the corresponding sentence embedding is defined as

$$e_{g_j} = f_{\mathrm{SBERT}}(g_j), \tag{10}$$

where $f_{\mathrm{SBERT}}(\cdot)$ denotes the Sentence-BERT embedding function [21].

Similarly, the embedding of the ith retrieved document is obtained as

$$e_{r_i} = f_{\mathrm{SBERT}}(r_i). \tag{11}$$

The semantic alignment between the jth generated sentence and the ith retrieved document is quantified using cosine similarity as

$$A_{ij} = \frac{e_{g_j} \cdot e_{r_i}}{\|e_{g_j}\| \|e_{r_i}\|}. \tag{12}$$

where A_{ij} denotes the evidence attribution score between the jth generated sentence and the ith retrieved document. Higher values indicate stronger semantic alignment between the generated sentence and the corresponding retrieved evidence.

For each generated sentence, the retrieved documents are ranked according to their evidence attribution scores as

$$\mathcal{R}_j^* = \mathrm{Sort}\left(A_{1j}, A_{2j}, \ldots, A_{kj}\right), \tag{13}$$

where $\mathcal{R}_j^*$ represents the ranked evidence set associated with the jth generated sentence.

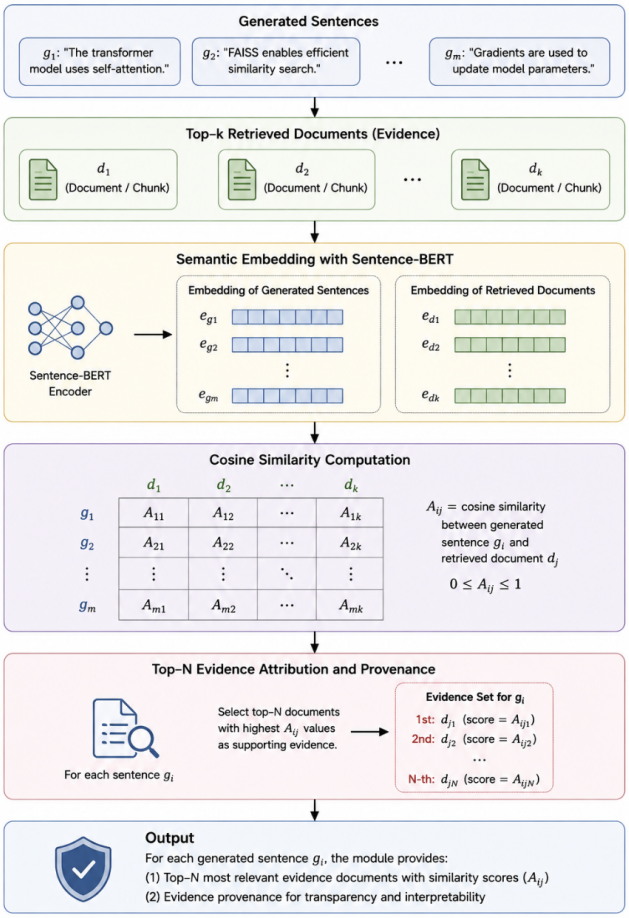


FIGURE 2. Workflow of the Explainable Evidence Attribution Module in the TrustRAG framework.

In addition to evidence attribution, the framework maintains provenance information by preserving the correspondence between each generated sentence and its supporting retrieved documents. This enables users to trace the evidence associated with individual generated statements, thereby improving transparency and facilitating evidence-based validation.

The ranked evidence set produced by the Explainable Evidence Attribution Module is subsequently forwarded to the Multi-Signal Hallucination Detection Module, where logical consistency, contextual semantic similarity, and embedding-level alignment are jointly evaluated to determine the reliability of each generated sentence.

As illustrated in Fig. 2, the Explainable Evidence Attribution Module establishes a transparent relationship between the generated sentences and the retrieved evidence.

D. MULTI-SIGNAL HALLUCINATION DETECTION MODULE

Although Retrieval-Augmented Generation (RAG) significantly improves factual grounding by incorporating external knowledge, retrieved evidence alone cannot guarantee the correctness of generated responses. Language models may still produce unsupported statements due to incomplete retrieval, semantic ambiguity, reasoning errors, or imperfect evidence utilization. To improve response reliability, the TrustRAG framework employs a multi-signal hallucination detection module that evaluates each generated sentence using logical consistency, contextual semantic similarity, and embedding-level alignment with the retrieved evidence.

Let the generated response be decomposed into m sentences as

$$G = \{g_1, g_2, \ldots, g_m\}, \tag{14}$$

where g_j denotes the jth generated sentence. Each sentence is verified independently against the retrieved evidence set $\mathcal{R}$.

Unlike approaches that rely on a single verification criterion, the proposed framework combines Natural Language Inference (NLI), BERTScore, and embedding-space cosine similarity to obtain complementary evidence regarding the reliability of each generated sentence. The integration of multiple verification signals reduces the limitations associated with individual evaluation metrics and provides a more robust basis for subsequent Trust Score computation.

1) Natural Language Inference Verification

Natural Language Inference determines whether a generated sentence is logically supported by the retrieved evidence. For each generated sentence g_j, the NLI model evaluates the logical relationship between the retrieved evidence and the generated sentence as

$$L_{\mathrm{NLI},j} = f_{\mathrm{NLI}}(\mathcal{R}, g_j), \tag{15}$$

where $f_{\mathrm{NLI}}(\cdot)$ denotes the Natural Language Inference model.

The NLI model assigns one of three logical relations:

$$L_{\mathrm{NLI},j} \in \{\mathrm{Entailment}, \mathrm{Neutral}, \mathrm{Contradiction}\}. \tag{16}$$

The entailment probability produced by the NLI model is used as the normalized NLI support score:

$$S_{\mathrm{NLI},j} \in [0, 1], \tag{17}$$

where higher values indicate stronger logical support for the generated sentence from the retrieved evidence.

2) Semantic Similarity using BERTScore

While NLI evaluates logical consistency, BERTScore measures contextual semantic similarity between each generated sentence and the retrieved evidence using contextual token embeddings. For the jth generated sentence, the semantic similarity score is defined as

$$S_{\mathrm{BERT},j} = \mathrm{BERTScore}(\mathcal{R}, g_j), \tag{18}$$

where

$$S_{\mathrm{BERT},j} \in [0, 1]. \tag{19}$$

Higher values indicate stronger contextual semantic correspondence between the generated sentence and the supporting evidence.

3) Embedding-Level Similarity

To further evaluate semantic consistency, Sentence-BERT embeddings are employed to compute cosine similarity between each generated sentence embedding and the aggregated embedding of the retrieved evidence.

The embedding of the jth generated sentence is defined as

$$e_{g_j} = f_{\mathrm{SBERT}}(g_j), \tag{20}$$

where $f_{\mathrm{SBERT}}(\cdot)$ denotes the Sentence-BERT embedding function.

The aggregated embedding of the retrieved evidence is computed as

$$e_{\mathcal{R}} = \frac{1}{k}\sum_{i=1}^{k} e_{r_i}, \tag{21}$$

where e_{r_i} represents the embedding of the ith retrieved document and k denotes the number of retrieved documents.

The embedding-level semantic similarity for the jth generated sentence is calculated using cosine similarity as

$$S_{\mathrm{COS},j} = \frac{e_{g_j} \cdot e_{\mathcal{R}}}{\|e_{g_j}\|\|e_{\mathcal{R}}\|}. \tag{22}$$

Higher cosine similarity values indicate stronger embedding-level semantic alignment between the generated sentence and the retrieved evidence.

The three verification scores $S_{\mathrm{NLI},j}$, $S_{\mathrm{BERT},j}$, and $S_{\mathrm{COS},j}$ provide complementary perspectives for assessing the reliability of the jth generated sentence. Specifically, NLI evaluates logical entailment, BERTScore measures contextual semantic correspondence, and cosine similarity quantifies embedding-space alignment. These scores are subsequently integrated by the Trust Score Computation Module to obtain a sentence-level reliability measure.

As illustrated in Fig. 3, the Multi-Signal Hallucination Detection Module evaluates each generated sentence using three complementary verification mechanisms. The NLI module determines logical consistency with the retrieved evidence, BERTScore measures contextual semantic similarity, and cosine similarity quantifies embedding-level semantic alignment. The resulting verification scores form a sentence-level verification vector, which is subsequently forwarded to the Trust Score Computation Module for reliability assessment.

E. TRUST SCORE COMPUTATION

The verification scores obtained from the Multi-Signal Hallucination Detection Module provide complementary evidence regarding the reliability of each generated sentence. Since each verification metric captures a different aspect of response correctness, the three signals are integrated into a unified Trust Score.

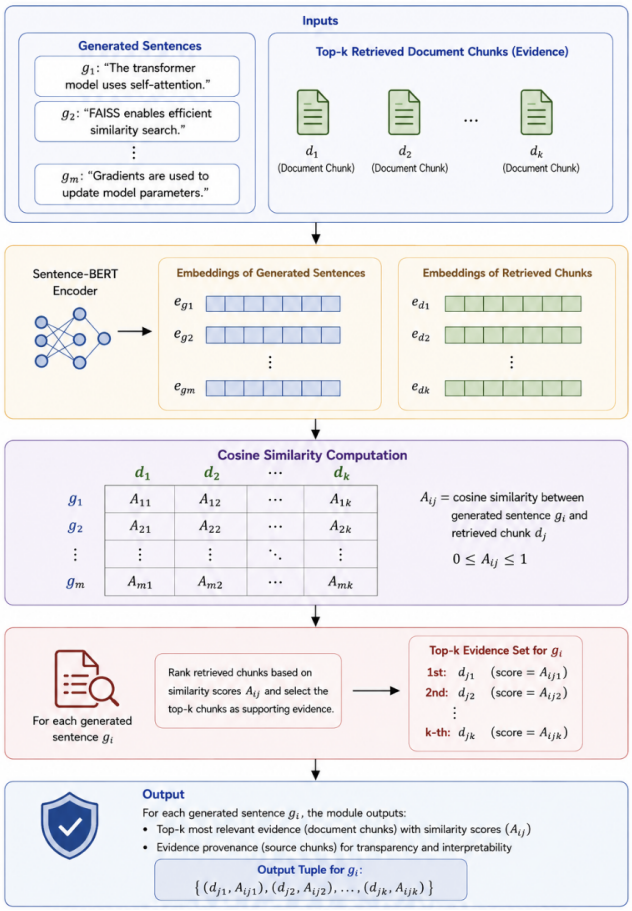


FIGURE 3. Workflow of the proposed Multi-Signal Hallucination Detection Module in the TrustRAG framework.

For the jth generated sentence, let

$$S_{\mathrm{NLI},j}, S_{\mathrm{BERT},j}, S_{\mathrm{COS},j} \in [0,1] \tag{23}$$

denote the normalized verification scores. The verification score vector is represented as

$$\mathbf{S}_j = [S_{\mathrm{NLI},j}, S_{\mathrm{BERT},j}, S_{\mathrm{COS},j}]. \tag{24}$$

The sentence-level Trust Score is computed using a weighted aggregation of the three verification signals:

$$T_j = w_1 S_{\mathrm{NLI},j} + w_2 S_{\mathrm{BERT},j} + w_3 S_{\mathrm{COS},j}, \tag{25}$$

subject to

$$w_1 + w_2 + w_3 = 1, \qquad w_i \ge 0, \quad i \in \{1,2,3\}. \tag{26}$$

Here, w_1, w_2, and w_3 represent the relative contributions of logical consistency, contextual semantic similarity, and embedding-level semantic alignment, respectively.

Since all verification scores are normalized to the interval $[0,1]$ and the weights are non-negative and sum to one, the resulting Trust Score satisfies

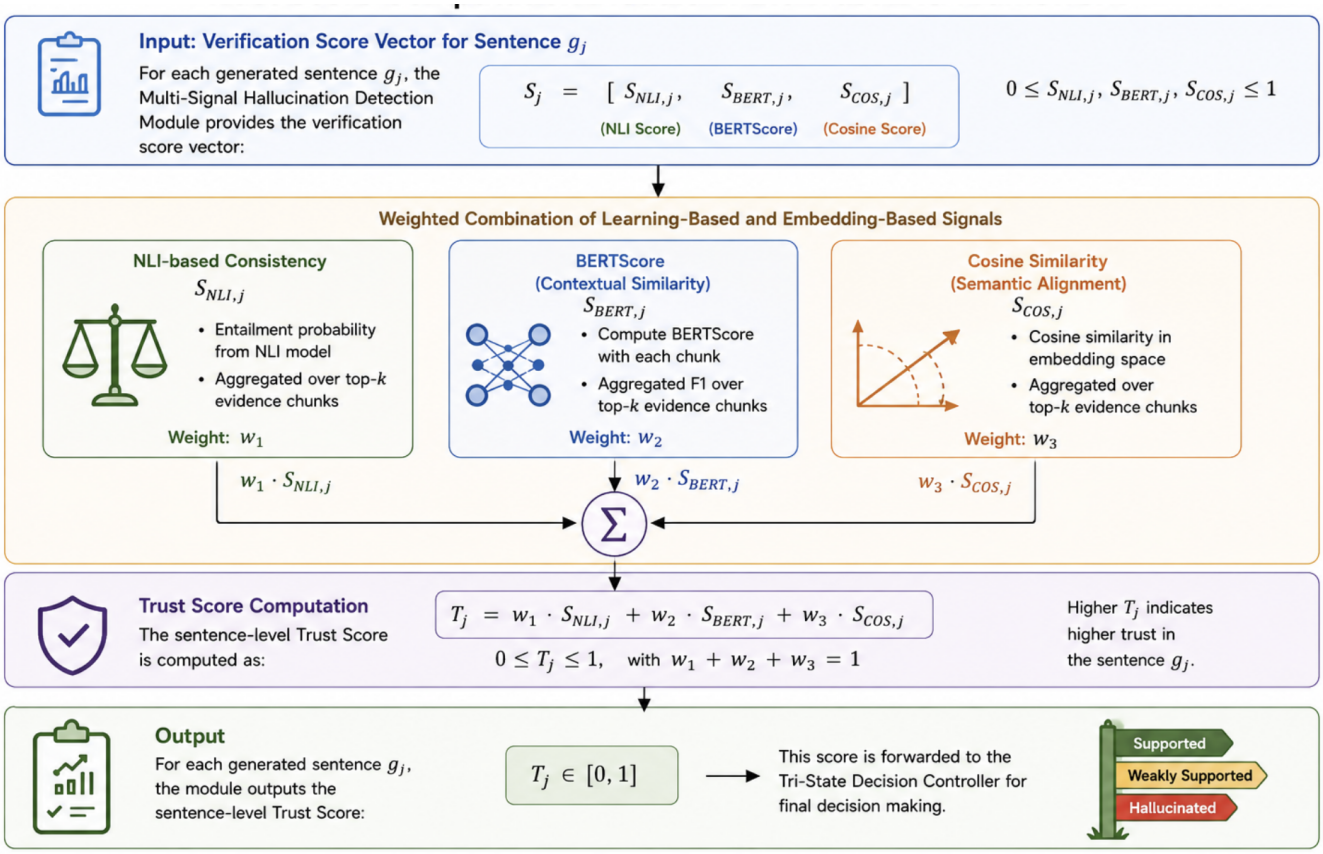


FIGURE 4. Workflow of the Trust Score Computation Module in the TrustRAG framework.

$$0 \le T_j \le 1. \tag{27}$$

The weighting function enables the model to achieve a balance in the combination of complementary signals for validation while compensating for the shortcomings of each metric. As such, the final Trust Score provides an aggregated measure of how reliable the generated sentence is and serves as an input into the Tri-State Decision Controller.

High Trust Score indicates agreement between the found evidence and the sentence that has been generated, while low scores indicate a lack of sufficient evidence and increased likelihood of hallucination.

As illustrated in Fig. 4, the Trust Score Computation Module integrates the three normalized verification scores obtained from the Multi-Signal Hallucination Detection Module. The weighted aggregation produces a sentence-level Trust Score in the range $[0, 1]$, which is subsequently forwarded to the Tri-State Decision Controller for sentence-level response validation.

F. TRI-STATE DECISION CONTROLLER

The sentence-level Trust Score provides a continuous measure of the reliability of each generated sentence. However, a continuous score alone does not provide an interpretable decision regarding whether a sentence is sufficiently supported by the retrieved evidence. To address this limitation, the TrustRAG framework incorporates a Tri-State Decision Controller that converts the sentence-level Trust Score into three interpretable states: *Supported*, *Weakly Supported*, and *Hallucinated*.

Let T_j denote the Trust Score associated with the jth generated sentence. Two decision thresholds are used to distinguish the three states. Based on the experimental configuration, the upper threshold is set to

$$\tau_H = 0.75, \tag{28}$$

and the lower threshold is set to

$$\tau_L = 0.50. \tag{29}$$

The sentence-level decision state is determined according to

$$D(g_j) = \begin{cases} \text{Supported}, & T_j \ge \tau_H, \\ \text{Weakly Supported}, & \tau_L \le T_j < \tau_H, \\ \text{Hallucinated}, & T_j < \tau_L. \end{cases} \tag{30}$$

Thus, sentences with Trust Scores of at least 0.75 are classified as Supported, sentences with Trust Scores between 0.50 and below 0.75 are classified as Weakly Supported,

and sentences with Trust Scores below 0.50 are classified as Hallucinated.

At the response level, the sentence-level decisions are aggregated to determine whether the complete generated response should be retained or suppressed. Let N_{hall} denote the number of generated sentences classified as Hallucinated and let N_{sent} denote the total number of generated sentences. The hallucination ratio is defined as

$$HR = \frac{N_{\mathrm{hall}}}{N_{\mathrm{sent}}}. \tag{31}$$

If more than 50% of the generated sentences are classified as Hallucinated, i.e.,

$$HR > 0.50, \tag{32}$$

the generated response is suppressed and the verified supporting evidence is returned instead. Otherwise, the generated response is retained together with its provenance information and supporting evidence.

A sentence classified as Supported exhibits strong agreement between the generated content and the retrieved evidence across the verification signals. Such sentences can be retained as reliable components of the generated response together with their associated evidence provenance.

The Weakly Supported state represents sentences for which the available evidence provides moderate support. These sentences may contain information that is semantically related to the retrieved evidence but do not achieve the confidence level required for direct acceptance. Therefore, they can be flagged for additional verification or presented with an appropriate uncertainty indication.

The Hallucinated state is assigned when the sentence-level Trust Score falls below the lower threshold. This indicates insufficient agreement between the generated sentence and the retrieved evidence and suggests an elevated risk of unsupported or factually inconsistent content.

The tri-state formulation provides a more informative decision mechanism than binary hallucination classification because it explicitly represents an intermediate confidence region. This is particularly useful in practical RAG applications where evidence may be incomplete or partially relevant and where forcing every generated sentence into a binary correct/incorrect category may result in inappropriate decisions.

The three states *Supported*, *Weakly Supported*, and *Hallucinated* constitute the sentence-level Trust Score classification. The *Blocked* outcome is a response-level suppression action rather than a fourth Trust Score classification state. It is activated when the hallucination ratio exceeds the predefined suppression criterion.

The complete sentence-level verification process can therefore be represented as

$$\mathcal{V}(g_j) = \{\mathcal{R}, g_j, \mathbf{S}_j, T_j, D(g_j)\}, \tag{33}$$

where $\mathcal{R}$ denotes the retrieved evidence, g_j represents the jth generated sentence, $\mathbf{S}_j$ denotes its verification score vector, T_j is the corresponding Trust Score, and $D(g_j)$ represents the resulting sentence-level decision state.

At the response level, the sentence-level decisions are aggregated using the hallucination ratio *HR* to determine whether the complete generated response is retained or suppressed. This decision mechanism completes the TrustRAG verification pipeline by transforming multiple evidence-based verification signals into an interpretable response-level decision.

G. COMPUTATIONAL COMPLEXITY ANALYSIS

The computational complexity of the proposed TrustRAG framework is determined by the major processing stages, including semantic embedding generation, vector-based retrieval, evidence attribution, multi-signal verification, and Trust Score computation. Let N denote the number of documents in the local knowledge repository, k the number of retrieved documents, m the number of generated sentences, and d the dimensionality of the dense embedding space.

The document embeddings are generated offline using the Sentence-BERT encoder and stored in the FAISS vector index. Since document embeddings can be computed once and reused for subsequent queries, this preprocessing operation does not contribute directly to the online query-processing cost. For N documents, the embedding generation cost is approximately

$$O(NC_{\mathrm{SBERT}}), \tag{34}$$

where C_{SBERT} represents the computational cost of encoding one document using Sentence-BERT.

During online processing, the input query is encoded using Sentence-BERT with a cost of

$$O(C_{\mathrm{SBERT}}). \tag{35}$$

The resulting query embedding is then used to retrieve the top-k documents from the FAISS index. For an exact similarity search over N dense vectors, the similarity computation requires

$$O(Nd). \tag{36}$$

However, FAISS supports efficient approximate nearest-neighbor indexing, substantially reducing the practical search cost for large document collections. The exact retrieval complexity therefore depends on the selected FAISS index structure and its associated search parameters.

The Explainable Evidence Attribution Module computes semantic alignment between each generated sentence and the k retrieved documents. Assuming that the corresponding embeddings have already been generated, the sentence-level cosine similarity operations require

$$O(mkd), \tag{37}$$

where m denotes the number of generated sentences and d denotes the embedding dimensionality.

The Multi-Signal Hallucination Detection Module evaluates each generated sentence using NLI, BERTScore, and cosine similarity. The NLI and BERTScore computations depend primarily on the lengths of the evidence and generated sentences and the underlying transformer architectures. Therefore, their model-dependent inference costs are represented as

$$O(C_{\mathrm{NLI}}) \tag{38}$$

and

$$O(C_{\mathrm{BERTScore}}), \tag{39}$$

respectively, where C_{NLI} and $C_{\mathrm{BERTScore}}$ denote the corresponding model-specific inference costs.

Since these verification operations are performed for each generated sentence, the overall model-dependent verification cost can be represented as

$$O\left(m(C_{\mathrm{NLI}} + C_{\mathrm{BERTScore}})\right). \tag{40}$$

The cosine similarity component of the verification stage requires $O(mkd)$ operations when each generated sentence is compared with the embeddings of the k retrieved documents.

Finally, the Trust Score computation combines three normalized verification scores using a constant number of arithmetic operations and therefore requires

$$O(1) \tag{41}$$

per generated sentence. Similarly, the Tri-State Decision Controller compares the sentence-level Trust Score against two predefined thresholds and therefore operates in constant time,

$$O(1) \tag{42}$$

per generated sentence.

Accordingly, excluding the model-dependent inference costs of the language and verification models, the dominant online computational components are query embedding generation, vector retrieval, and sentence-level evidence similarity computation. The use of a precomputed local vector index and top-k retrieval limits the amount of evidence processed by the subsequent verification modules.

Although the verification cost depends on the number of generated sentences, the retrieval stage restricts the evidence set to the selected top-k documents. This design supports a scalable verification pipeline while enabling offline deployment in privacy-sensitive environments.

IV. EXPERIMENTAL SETUP

A. EXPERIMENTAL ENVIRONMENT AND IMPLEMENTATION

The proposed TrustRAG framework was implemented using open-source software to support offline and privacy-preserving deployment. All retrieval, response generation, evidence attribution, and hallucination verification components were executed locally without relying on external application programming interfaces (APIs) or cloud-based inference services. This configuration ensures that the documents used by the system remain within the local computational environment throughout the retrieval and response-generation process. The overall implementation was developed in Python and integrates semantic embedding, vector retrieval, local language-model inference, and multi-signal response verification within a unified processing pipeline.

The knowledge base was constructed from approximately 250 domain-specific technical documents covering enterprise policy manuals, engineering standards, and software documentation. The source documents were converted into machine-readable text and divided into overlapping segments of 512 tokens with an overlap of 64 tokens. This preprocessing procedure resulted in approximately 9,800 indexed text chunks, which formed the offline retrieval corpus. The resulting chunks were transformed into dense semantic representations using the all-MiniLM-L6-v2 Sentence Transformer and indexed in a FAISS vector database for efficient nearest-neighbor retrieval.

For response generation, a locally hosted Llama 3 8B Instruct model was deployed through the Ollama inference framework. Given a user query, the retrieval module identifies the most relevant evidence chunks and supplies them as contextual information to the local language model. Each generated sentence is subsequently evaluated using Natural Language Inference (NLI), BERTScore, and embedding-space cosine similarity. The resulting verification signals are integrated into the Trust Score mechanism described in Section III.

The complete experimental configuration therefore maintains the retrieval, generation, and verification stages within a local execution environment. This design is particularly relevant for applications in which sensitive organizational information should not be transmitted to external cloud services.

B. KNOWLEDGE BASE AND QUERY BENCHMARK

The experimental evaluation was conducted using a manually curated benchmark designed to assess both evidence-grounded response generation and hallucination detection. The knowledge base contained approximately 250 domain-specific technical documents representing three primary categories: enterprise policy manuals, engineering standards, and software documentation. These documents were selected to provide heterogeneous factual content and to evaluate the ability of the proposed framework to retrieve and utilize evidence from different technical contexts.

Following document preprocessing and segmentation, approximately 9,800 text chunks were indexed in the local FAISS vector database. The retrieval system was evaluated using a benchmark of 400 manually curated natural-language queries. The benchmark consisted of 250 answerable queries and 150 hallucination-inducing queries. Answerable queries

were designed to have sufficient supporting information within the knowledge base, whereas hallucination-inducing queries were constructed to test the ability of the framework to identify responses that could not be adequately supported by the available evidence.

For each query, the system retrieved the top-ranked evidence chunks from the local vector database and supplied the retrieved context to the locally hosted Llama 3 8B Instruct model. The generated response was subsequently decomposed into individual sentences for sentence-level verification. Each sentence was evaluated using the three verification signals defined in the proposed framework, namely NLI entailment, BERTScore, and embedding-space cosine similarity.

The benchmark was designed to evaluate the framework from multiple perspectives, including hallucination classification, retrieval effectiveness, response quality, and evidence provenance. This evaluation protocol enables the individual contributions of retrieval, verification, and trust-aware decision making to be examined systematically.

C. EVALUATION METRICS

The performance of TrustRAG was evaluated using complementary metrics covering hallucination detection, retrieval effectiveness, response quality, and evidence provenance.

For hallucination detection, Precision, Recall, F1-score, and Area Under the Receiver Operating Characteristic Curve (AUC-ROC) were used. Precision measures the proportion of responses identified as hallucinated that are actually hallucinated, whereas Recall measures the ability to identify the hallucinated responses present in the benchmark. The F1-score provides a harmonic mean of Precision and Recall, while AUC-ROC evaluates the discrimination capability of the verification framework across different decision thresholds.

Retrieval effectiveness was evaluated using Recall@5, Mean Reciprocal Rank (MRR), and Normalized Discounted Cumulative Gain at rank 5 (NDCG@5). Recall@5 measures whether relevant evidence is retrieved among the five highest-ranked results. MRR evaluates the reciprocal rank of the first relevant retrieved item, while NDCG@5 accounts for the ranking position of relevant evidence within the top-five results.

Response–evidence semantic alignment was evaluated using BERTScore F1, which measures contextual semantic similarity between the generated response and its supporting evidence. In addition, a provenance score was used to quantify the degree to which generated content could be associated with retrieved supporting evidence.

The evaluation therefore considers both the retrieval and generation stages as well as the downstream verification process, providing a comprehensive assessment of the proposed TrustRAG framework.

D. EXPERIMENTAL PROTOCOL

The experimental protocol follows the complete TrustRAG processing pipeline. For each query, the system first performs dense semantic retrieval from the offline FAISS index and selects the top-ranked evidence chunks. The retrieved evidence is then provided to the locally hosted language model for response generation. The generated response is subsequently evaluated at the sentence level using the NLI, BERTScore, and cosine similarity verification mechanisms.

For each generated sentence s_j, the three verification scores are computed and integrated using the weighted aggregation scheme defined in Section III. The resulting Trust Score is used by the tri-state decision controller to classify the sentence as Supported, Weakly Supported, or Hallucinated. The thresholds used in the experimental evaluation are 0.75 and 0.50, respectively. Sentences with Trust Scores of at least 0.75 are classified as Supported, those with scores from 0.50 to below 0.75 are classified as Weakly Supported, and those with scores below 0.50 are classified as Hallucinated.

At the response level, if more than 50% of the generated sentences are classified as hallucinated, the response is suppressed and only the verified evidence is returned. Otherwise, the generated response is retained together with its provenance information and supporting evidence. This protocol enables the evaluation to measure not only sentence-level hallucination detection but also the practical response-level behavior of the proposed framework.

Three categories of experiments were subsequently conducted: overall hallucination detection and retrieval evaluation, tri-state decision analysis, and ablation analysis. In addition, TrustRAG was compared against a conventional RAG baseline without explicit hallucination verification to quantify the effectiveness of the proposed verification and trust-aware components.

V. RESULTS AND DISCUSSION

A. HALLUCINATION DETECTION AND RETRIEVAL PERFORMANCE

The results of the overall performance of the TrustRAG framework in terms of hallucination detection, retrieval, response generation, and provenance metrics are presented in Table 2. The TrustRAG framework demonstrates Precision of 0.912 and Recall of 0.884 and thereby obtains the F1-score of 0.898. The value of AUC-ROC is 0.951.

TABLE 2. Overall Performance of TrustRAG

Metric	Value
Precision	0.912
Recall	0.884
F1-score	0.898
AUC-ROC	0.951
Recall@5	0.934
MRR	0.871
NDCG@5	0.902
BERTScore F1	0.917
Provenance Score	0.936

In the experiment performed to detect hallucinations, we can conclude that multi-signal verification model provides balanced classification. In particular, an F1 score of 0.898 proves that the system retains good balance between detection of hallucinations and avoiding wrong classifications of non-hallucinations. An AUC-ROC score of 0.951 proves discrimination at different thresholds.

Furthermore, in the retrieval results, one can notice the effectiveness of detecting evidence. The value of Recall@5 is equal to 0.934 and shows that related evidence is found in the first five documents for the majority of queries. The value of MRR is 0.871 and shows that related evidence is positioned at the top of the list. Similarly, the NDCG@5 value of 0.902 indicates that the ranking of retrieved evidence remains effective when both relevance and retrieval position are considered.

The response–evidence semantic alignment results provide additional evidence of effective grounding. A BERTScore F1 of 0.917 indicates strong contextual semantic correspondence between the generated responses and their supporting evidence. The provenance score of 0.936 further demonstrates that generated responses can be associated with identifiable supporting evidence from the offline knowledge base. These results are particularly relevant to the objective of improving transparency, since response quality is evaluated together with evidence traceability rather than in isolation.

Overall, the results indicate that TrustRAG provides consistent performance across the three principal stages of the framework: evidence retrieval, response verification, and provenance-aware response generation. Taken together, the retrieval, hallucination detection, semantic alignment, and provenance results provide a broader assessment of response reliability than evaluating generated responses using generation quality alone.

B. TRI-STATE DECISION ANALYSIS

The effectiveness of the proposed tri-state decision controller was evaluated using both answerable and hallucination-inducing queries. The resulting classification distributions are presented in Table 3. The first three columns represent the sentence-level decision states of Supported, Weakly Supported, and Hallucinated. The Blocked column reports the percentage of responses that were suppressed at the response level when the hallucination ratio exceeded the predefined threshold.

TABLE 3. Tri-State Classification Distribution

Query Type	Supported	Weakly Supported	Hallucinated	Blocked
Answerable	81.0	12.5	6.5	2.4
Hallucination-inducing	22.1	19.2	58.7	41.3

Note: Supported, Weakly Supported, and Hallucinated represent the sentence-level classification distribution. Blocked denotes the percentage of complete responses suppressed at the response level when the hallucination ratio exceeds 0.50; therefore, Blocked is not included in the tri-state percentage sum.

As shown in Table 3, the majority of sentences generated for answerable queries are classified as Supported, accounting for 81.0% of the evaluated sentences. A further 12.5% are classified as Weakly Supported, indicating partial or moderate evidence support, while 6.5% are classified as Hallucinated.

In contrast, hallucination-inducing queries exhibit a substantially different classification pattern. The Hallucinated category accounts for 58.7% of the evaluated sentences, compared with 6.5% for answerable queries. This difference indicates that the Trust Score and the associated decision thresholds provide useful separation between evidence-supported and unsupported generated content.

The Supported category for hallucination-inducing queries is 22.1%, while 19.2% of the responses are classified as Weakly Supported. These cases indicate that the presence of a hallucination-inducing query does not necessarily result in an unsupported response when relevant evidence can still be retrieved from the knowledge base. Consequently, the decision controller evaluates the generated content against the available evidence rather than classifying the query solely according to its predefined category.

The response suppression mechanism provides an additional safeguard against unreliable generation. According to the decision policy, a response containing hallucinated sentences exceeding the specified proportion is suppressed and the corresponding verified evidence is returned instead. This mechanism prevents responses with a high concentration of unsupported content from being presented as reliable information. The Blocked percentages are reported separately from the sentence-level classification distribution because blocking is a response-level safety action rather than an additional tri-state classification.

Overall, the tri-state analysis demonstrates the value of retaining an intermediate Weakly Supported state rather than using a binary Supported–Hallucinated classification. The intermediate category allows sentences with partial evidence support to be distinguished from strongly grounded sentences and high-risk hallucinated sentences. This provides a more informative basis for response-level reliability assessment and supports evidence-aware decision making.

C. ABLATION STUDY

An ablation study was conducted to examine the individual and combined contributions of the verification components used in TrustRAG. The evaluated configurations include NLI-only verification, BERTScore-only verification, cosine-similarity-only verification, a combination of NLI and BERTScore, and the complete TrustRAG configuration. The corresponding results are presented in Table 4.

The results in Table 4 demonstrate a progressive improvement as complementary verification signals are incorporated. Among the individual verification mechanisms, NLI achieves

TABLE 4. Ablation Study of the Verification Components

Configuration	Precision	Recall	F1	AUC
NLI Only	0.871	0.842	0.856	0.918
BERTScore Only	0.803	0.781	0.792	0.874
Cosine Only	0.742	0.716	0.729	0.821
NLI+BERTScore	0.896	0.861	0.878	0.939
TrustRAG	0.912	0.884	0.898	0.951

the highest performance, with a Precision of 0.871, Recall of 0.842, F1-score of 0.856, and AUC of 0.918. This indicates that logical entailment provides a strong signal for distinguishing evidence-supported content from hallucinated content.

BERTScore-only verification achieves a Precision of 0.803, Recall of 0.781, F1-score of 0.792, and AUC of 0.874. Although these results are lower than those obtained using NLI, the performance demonstrates that contextual semantic similarity provides useful information for response verification.

Cosine-similarity-only verification produces the lowest performance among the individual configurations, with a Precision of 0.742, Recall of 0.716, F1-score of 0.729, and AUC of 0.821. This result indicates that embedding-level similarity alone is insufficient to reliably distinguish factual support from hallucinated content. Semantic closeness in the embedding space does not necessarily imply logical entailment or factual consistency.

Combining NLI with BERTScore improves the performance to a Precision of 0.896, Recall of 0.861, F1-score of 0.878, and AUC of 0.939. The improvement over the individual configurations demonstrates that logical consistency and contextual semantic similarity provide complementary information for response verification.

The complete TrustRAG configuration achieves the highest performance across all four metrics, with a Precision of 0.912, Recall of 0.884, F1-score of 0.898, and AUC of 0.951. Compared with the NLI-only configuration, the complete framework improves Precision by 4.1 percentage points, Recall by 4.2 percentage points, F1-score by 4.2 percentage points, and AUC by 3.3 percentage points. These improvements indicate that contextual semantic similarity and embedding-level semantic alignment provide complementary information beyond logical entailment alone.

The ablation results therefore support the use of multiple verification signals within TrustRAG. While NLI provides the strongest individual verification signal in this evaluation, the combination of NLI, BERTScore, and cosine similarity produces the highest overall classification performance. This observation supports the multi-signal design adopted for the Trust Score computation.

D. COMPARISON WITH CONVENTIONAL RAG

To evaluate the practical contribution of the proposed verification and trust-aware mechanisms, TrustRAG was compared with a conventional Retrieval-Augmented Generation (RAG) baseline. The baseline performs document retrieval followed by response generation without the explicit multi-signal hallucination detection, provenance-aware verification, and trust-based response control incorporated in TrustRAG. The comparative results are presented in Table 5.

TABLE 5. Comparison Between Conventional RAG and TrustRAG

Metric	Baseline RAG	TrustRAG
Hallucinated Sentences (%)	18.6	2.1
Provenance Score	0.741	0.936
Answer BERTScore	0.868	0.917
Latency (s)	6.2	9.8
Citation Coverage	0	1

As shown in Table 5, TrustRAG substantially reduces the proportion of hallucinated sentences compared with the conventional RAG baseline. The hallucinated-sentence rate decreases from 18.6% for the baseline to 2.1% for TrustRAG. This corresponds to a reduction of 16.5 percentage points, indicating the effectiveness of the explicit response verification mechanism in identifying unsupported generated content.

TrustRAG also achieves a higher provenance score of 0.936 compared with 0.741 for the baseline RAG system. The improvement indicates stronger association between generated responses and identifiable supporting evidence. This is consistent with the evidence attribution and provenance mechanisms incorporated into the TrustRAG pipeline.

The Answer BERTScore increases from 0.868 for the conventional RAG baseline to 0.917 for TrustRAG, indicating stronger response–evidence semantic alignment. This result suggests that the additional verification stages not only suppress unsupported content but also maintain a high degree of semantic alignment with the retrieved evidence.

The additional verification stages increase the average latency from 6.2 s for conventional RAG to 9.8 s for TrustRAG. This increase reflects the computational overhead associated with sentence-level NLI, BERTScore, cosine-similarity verification, and Trust Score computation. The additional latency represents a trade-off for improved hallucination control and evidence provenance. Citation Coverage is reported as a normalized proportion, where 0 indicates that no generated content is associated with an explicit supporting citation and 1 indicates complete citation coverage of the evaluated generated content.

Overall, the comparison demonstrates that TrustRAG substantially reduces hallucinated sentence generation while improving evidence provenance and response–evidence semantic alignment. The percentage of hallucinated sentences decreases from 18.6% for conventional RAG to 2.1% for TrustRAG, while the Provenance Score increases from 0.741 to 0.936 and the Answer BERTScore increases from 0.868 to 0.917. These improvements are achieved with an increase in average latency from 6.2 s to 9.8 s due to the additional verification and trust-aware processing stages. The results therefore indicate a practical trade-off between computational overhead and improved hallucination control, evidence provenance, and response–evidence semantic alignment.

E. OVERALL DISCUSSION AND KEY FINDINGS

The experimental results collectively demonstrate that TrustRAG provides an integrated approach to hallucination control, evidence grounding, provenance tracking, and trust-aware response validation. The hallucination detection results, together with the retrieval evaluation, indicate that reliable response verification depends on both effective evidence retrieval and accurate assessment of the relationship between generated content and retrieved evidence.

The ablation analysis further supports the multi-signal design of TrustRAG. Although NLI provides the strongest standalone verification performance, combining NLI, BERTScore, and cosine similarity achieves higher overall classification performance. It can be inferred that logic reasoning, semantic similarity, and alignment are complementary features when it comes to recognizing unsupported generated data.

As compared to traditional RAG approaches, the benefits of using evidence attribution and trust-aware verification in the process of generation can be observed. The introduction of TrustRAG decreases the amount of hallucinations in the output text, improves provenance, and increases the alignment between response and evidence. It can be said that all these facts demonstrate that the introduced approach can improve reliability without using only generation model to produce factual response.

It can be observed that the proposed solution requires some computation cost. TrustRAG takes more time for processing because the processes of evidence attribution, multi-signal verification, and trust-aware control happen before the delivery of the answer to the user. Though it is a question of latency, it brings benefits of better control over the generated data and more transparent evidence attribution.

Finally, the offline execution model represents an important deployment characteristic of TrustRAG. Retrieval, generation, and verification are performed locally without dependence on external cloud services or APIs. This makes the framework suitable for privacy-sensitive and air-gapped environments where local control of knowledge, inference, and evidence verification is required. Overall, the findings support the central design principle of integrating retrieval, explainability, and multi-signal verification within a unified offline RAG pipeline.

VI. CONCLUSION AND FUTURE WORK

This paper presented TrustRAG, an offline Retrieval-Augmented Generation framework designed to improve the reliability, transparency, and traceability of generated responses. The framework integrates dense semantic retrieval, explainable evidence attribution, local Large Language Model generation, and multi-signal hallucination detection within a unified pipeline. Natural Language Inference, BERTScore, and embedding-based cosine similarity are combined through a Trust Score mechanism, while a tri-state decision controller classifies each generated sentence as Supported, Weakly Supported, or Hallucinated before response-level reliability control is applied.

The experimental evaluation demonstrates the effectiveness of the framework across hallucination detection, retrieval, response–evidence semantic alignment, and evidence provenance. TrustRAG achieves an F1-score of 0.898 and an AUC-ROC of 0.951 for hallucination detection, while Recall@5 and NDCG@5 reach 0.934 and 0.902, respectively. The framework also achieves a BERTScore F1 of 0.917 and a Provenance Score of 0.936. The ablation analysis further demonstrates that combining NLI, BERTScore, and cosine similarity provides higher overall classification performance than the individual verification configurations.

The comparison with conventional RAG further demonstrates the practical effect of the proposed verification pipeline. The hallucinated-sentence rate decreases from 18.6% for conventional RAG to 2.1% for TrustRAG, while the Provenance Score increases from 0.741 to 0.936. There is a positive change in the value of the BERTScore metric for The Answer, which increased from 0.868 to 0.917. The increase in the value of the BERTScore metric is observed together with an increase in latency from 6.2 seconds to 9.8 seconds because of the overhead of evidence attribution, multi-signal verification, and trust-based response control.

An offline mode of operation of the TrustRAG framework allows deploying it in air-gapped and privacy-critical environments, where retrieval, generation, and verification cannot depend on any cloud services. In addition, with the help of the integration of evidence retrieval, provenance-aware explainability, sentence-level reliability estimation, and response-level control, the framework provides an all-in-one solution for deployment of a retrieval-augmented generation that can be trusted.

The future research will include developing adaptive methods of Trust Score weights and decision thresholds learning to decrease dependency on parameter setting. It also includes the research of multi-hop evidence aggregation for multi-document retrieval, multilingual retrieval, hallucination verification, and domain-specific adaptation of the framework to applications like healthcare, legal analytics, and enterprise knowledge management.

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